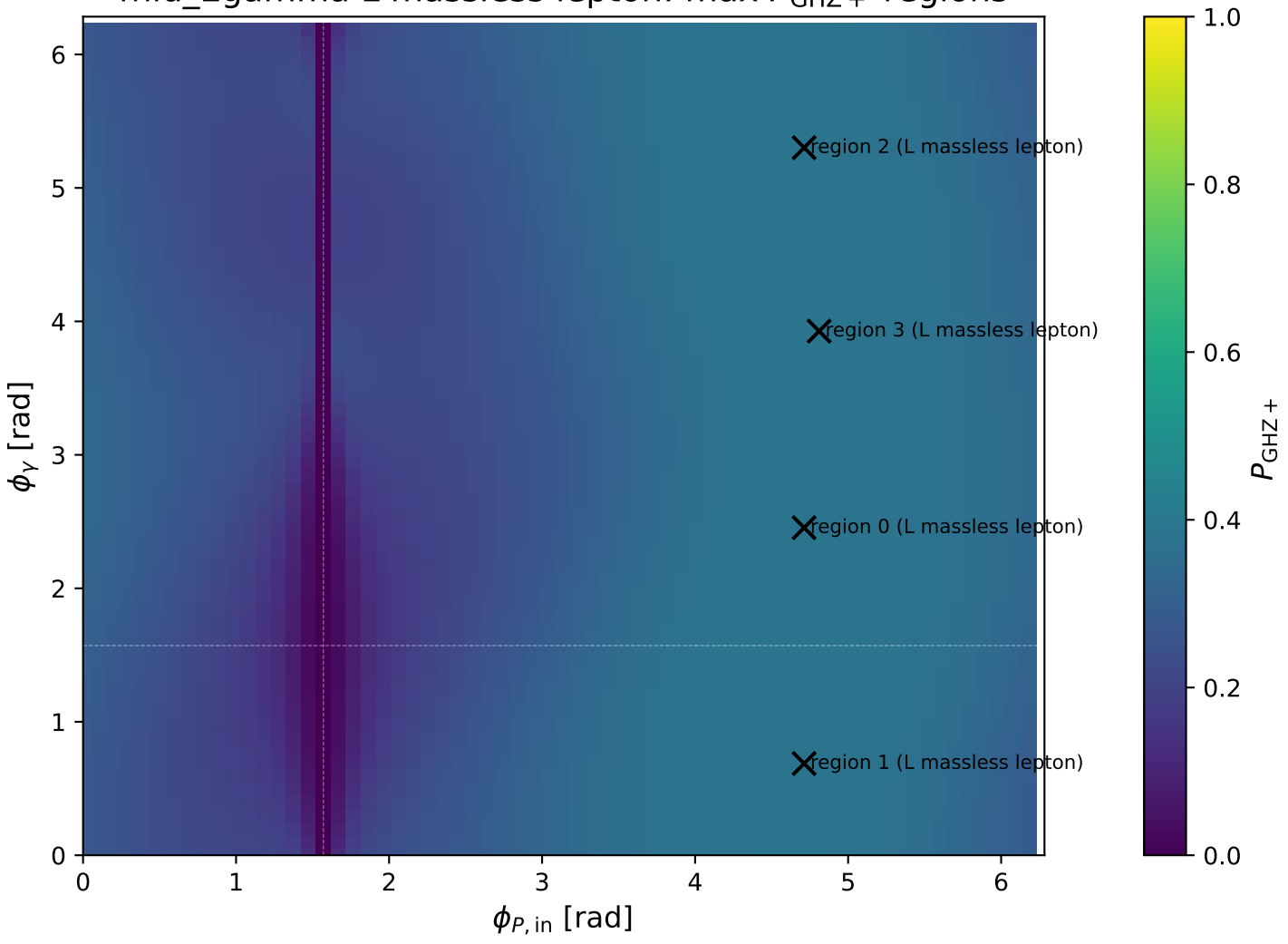
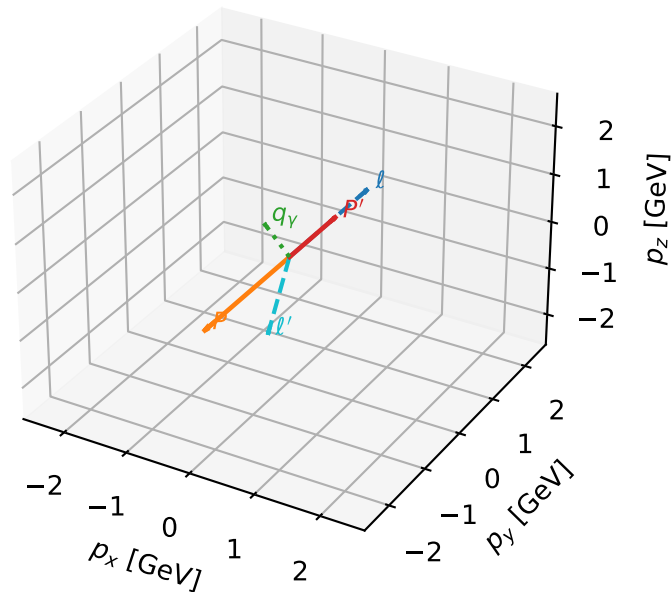


mid_Egamma L massless lepton: max $P_{\text{GHZ}+}$ regions



L massless lepton characteristic max P_{GHZ} + configuration

3D momenta

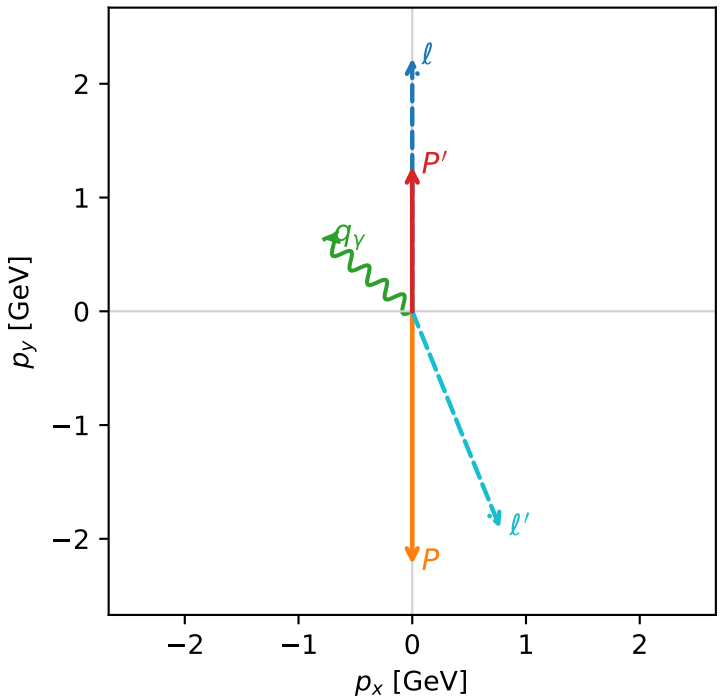


ghz_purity_mid_egamma_l_lepton_region_0 $P_{\text{GHZ}}=0.384124$
region: mid_Egamma_L_lepton_region_0
outgoing-state purity: 0.998238
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.27253$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_P=4.71239$
 $E_\gamma=1$, $\phi_\gamma=2.45437$
 $Q^2=17.6377$, $x_B=0.919354$, $t=-11.5344$, $W^2=2.42703$, $y=0.92968$
 $\theta(l', \gamma)=151.441$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 l' $E= 2.0576$ $p_x= 0.7730$ $p_y= -1.9069$ $p_z= 0.0000$
 P' $E= 1.5809$ $p_x= 0.0000$ $p_y= 1.2725$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

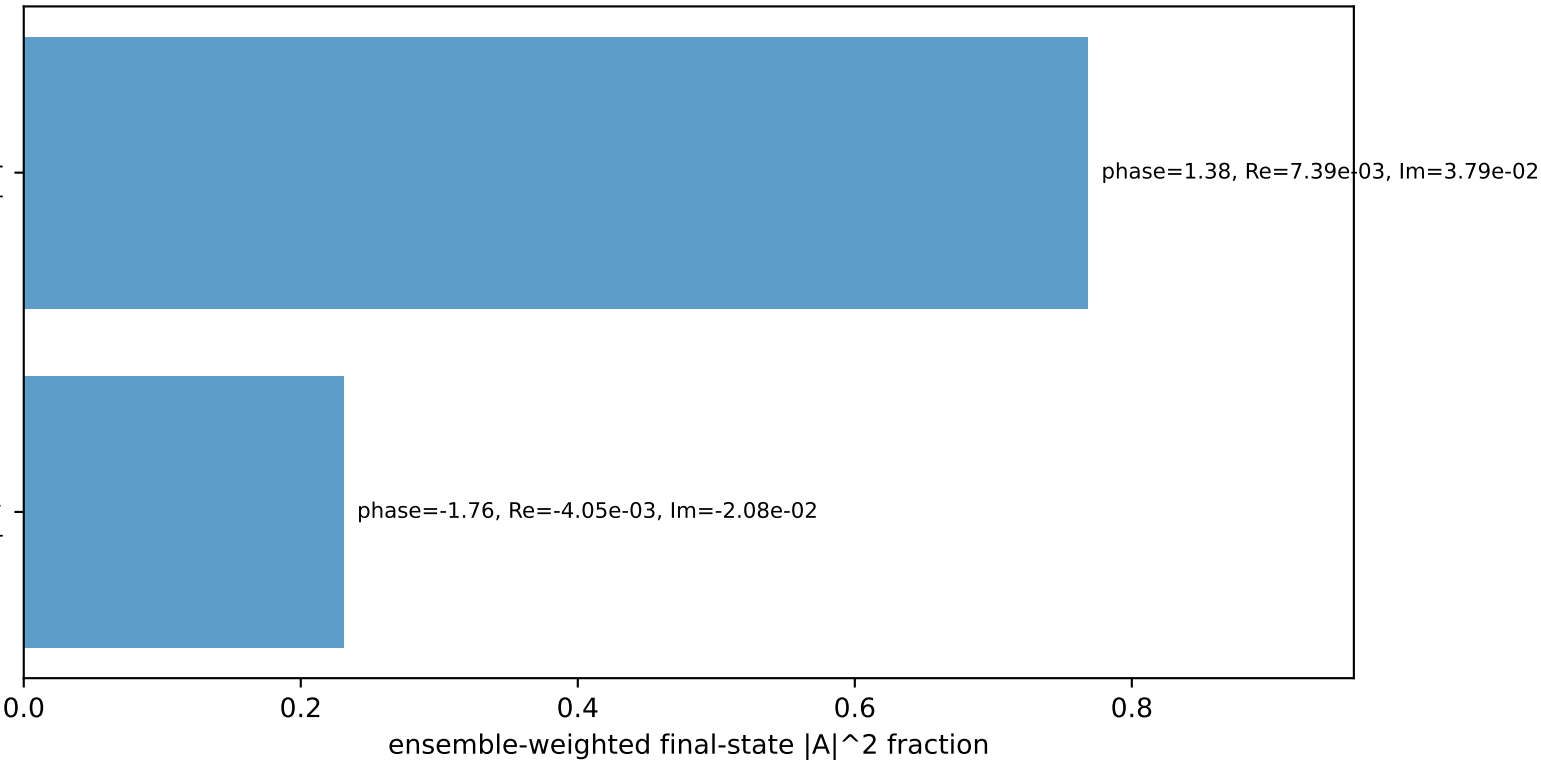
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton $h=+1$

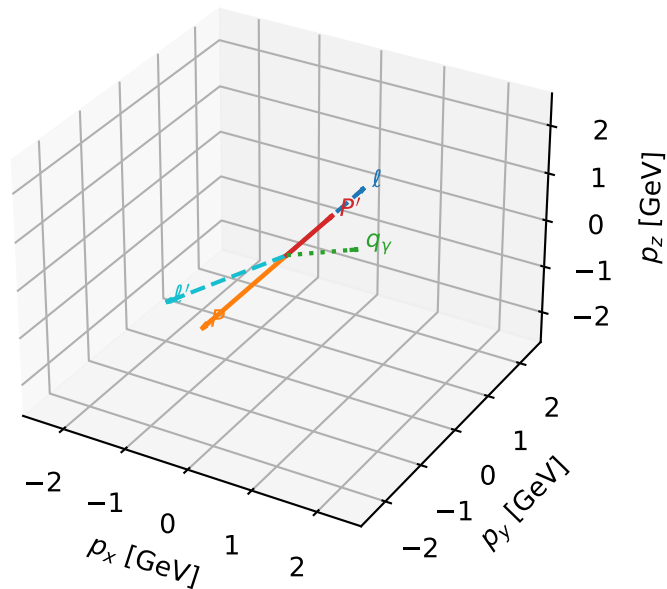
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton $h=+1$

Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton characteristic max P_{GHZ} + configuration

3D momenta

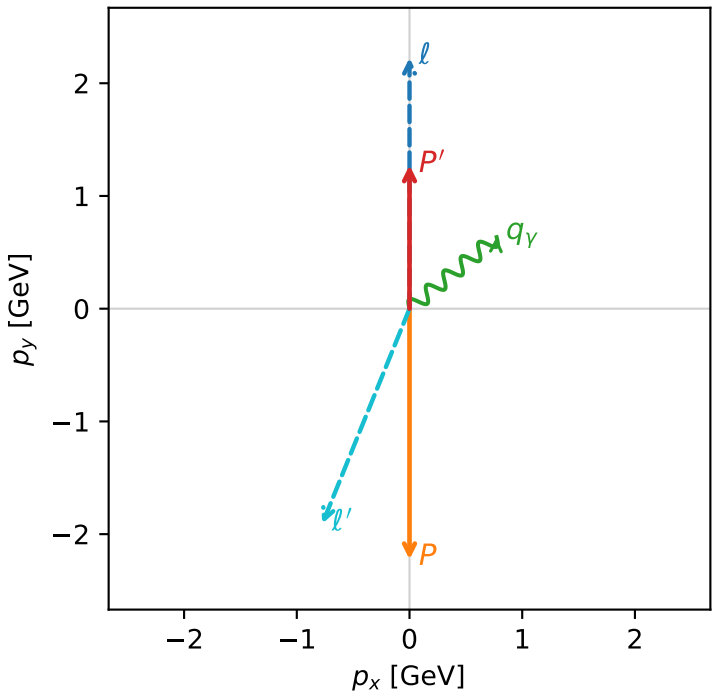


ghz_purity_mid_egamma_l_lepton_region_1 $P_{\text{GHZ}}=0.384124$
region: mid Egamma L lepton region_1
outgoing-state purity: 0.998238
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_{l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.27253$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_P=4.71239$
 $E_\gamma=1$, $\phi_\gamma=0.687223$
 $Q^2=17.6377$, $x_B=0.919354$, $t=-11.5344$, $W^2=2.42703$, $y=0.92968$
 $\theta(l', \gamma)=151.441$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 l' $E= 2.0576$ $p_x= -0.7730$ $p_y= -1.9069$ $p_z= 0.0000$
 P' $E= 1.5809$ $p_x= 0.0000$ $p_y= 1.2725$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

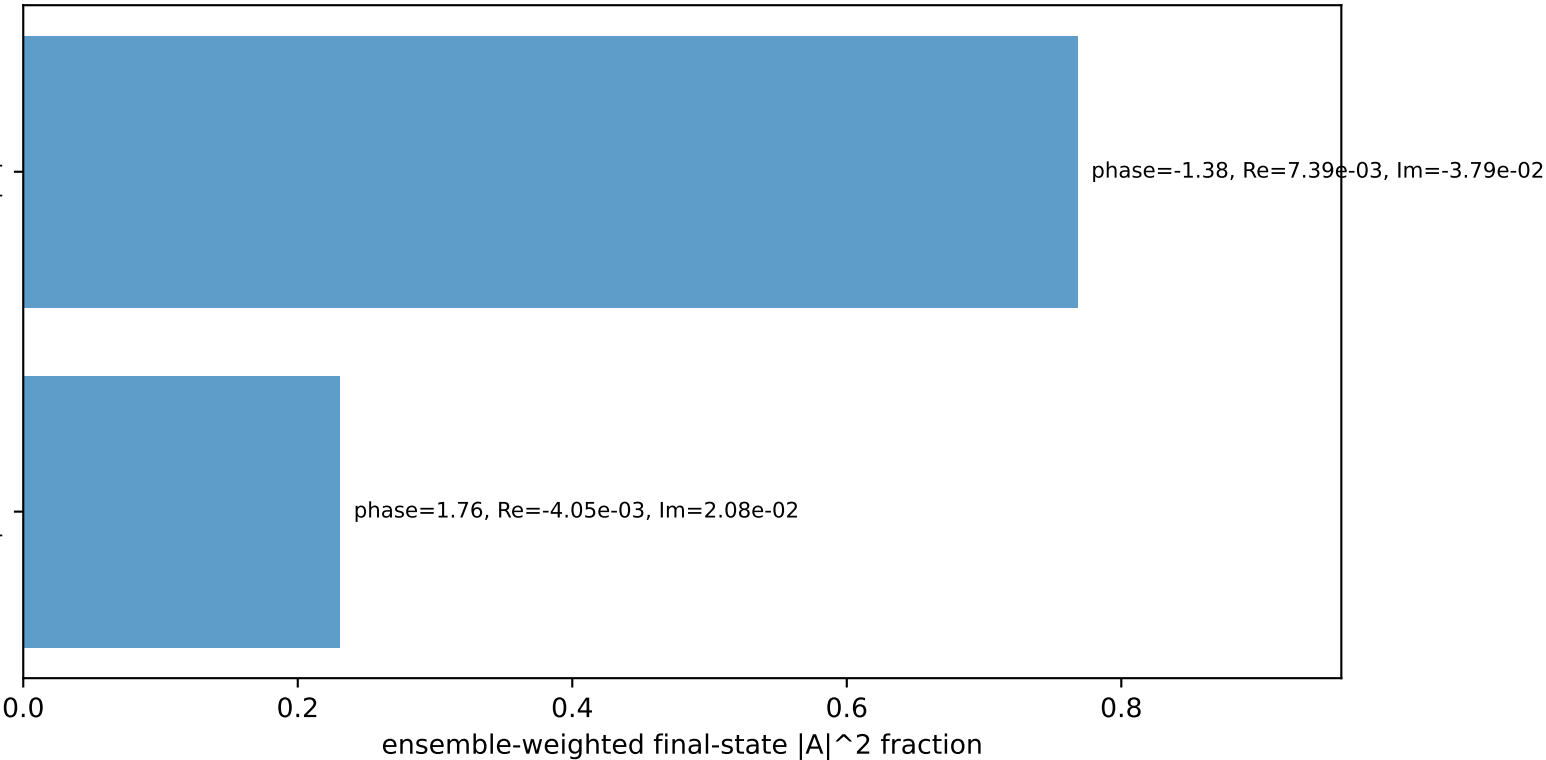
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton h=+1

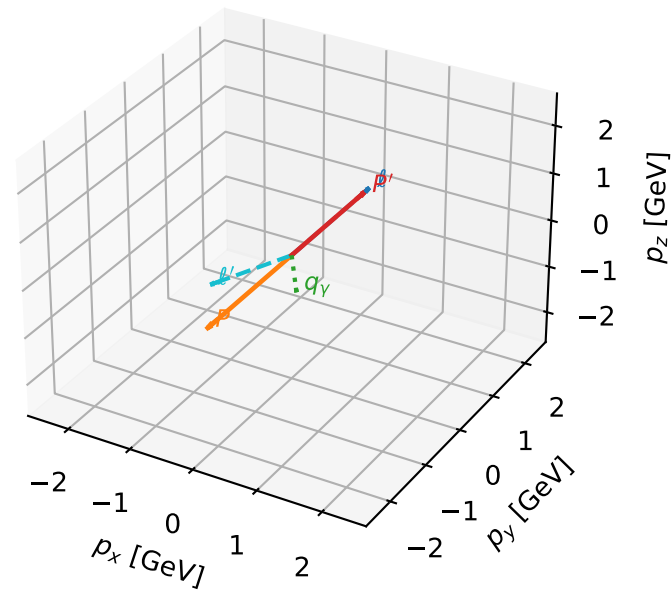
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton h=+1

Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton characteristic max P_{GHZ} + configuration

3D momenta

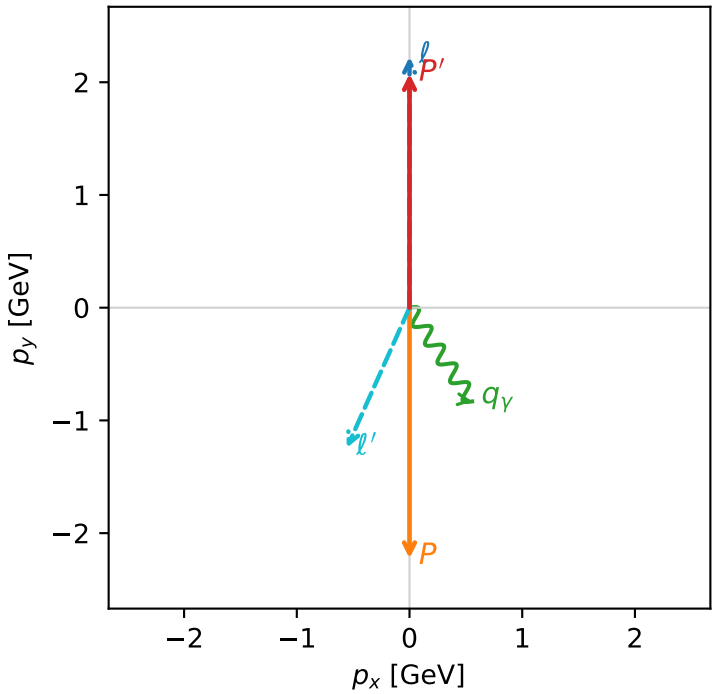


ghz_purity_mid_egamma_l_lepton_region_2 $P_{\text{GHZ}}=0.383886$
region: mid Egamma L lepton region_2
outgoing-state purity: 0.998883
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_{l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.07465$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_P=4.71239$
 $E_\gamma=1$, $\phi_\gamma=5.30144$
 $Q^2=11.5886$, $x_B=0.591487$, $t=-18.4631$, $W^2=8.88355$, $y=0.949422$
 $\theta(l', \gamma)=57.8296$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 l' $E= 1.3617$ $p_x= -0.5556$ $p_y= -1.2432$ $p_z= 0.0000$
 P' $E= 2.2768$ $p_x= 0.0000$ $p_y= 2.0746$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.5556$ $p_y= -0.8315$ $p_z= 0.0000$

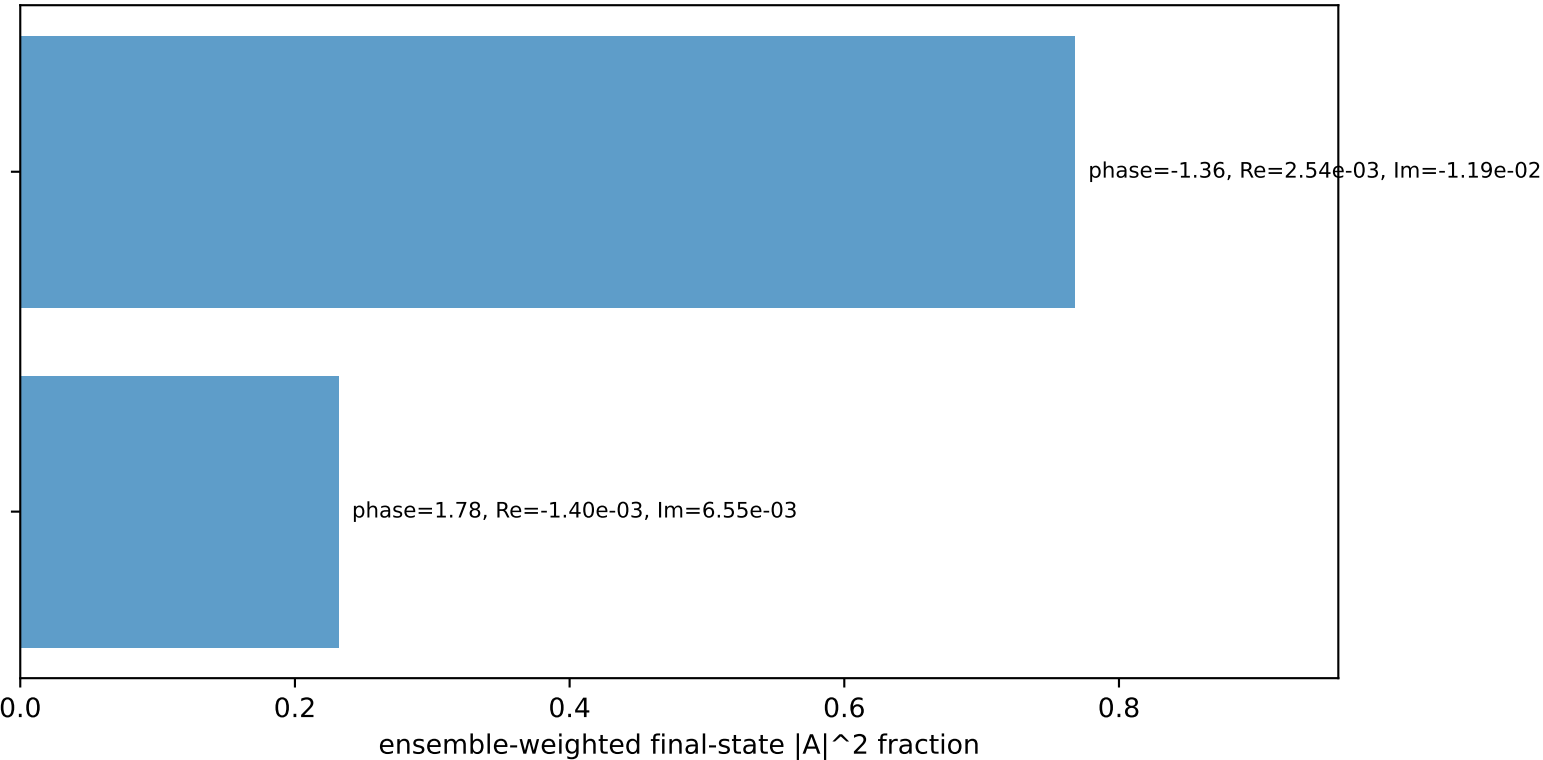
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton h=+1

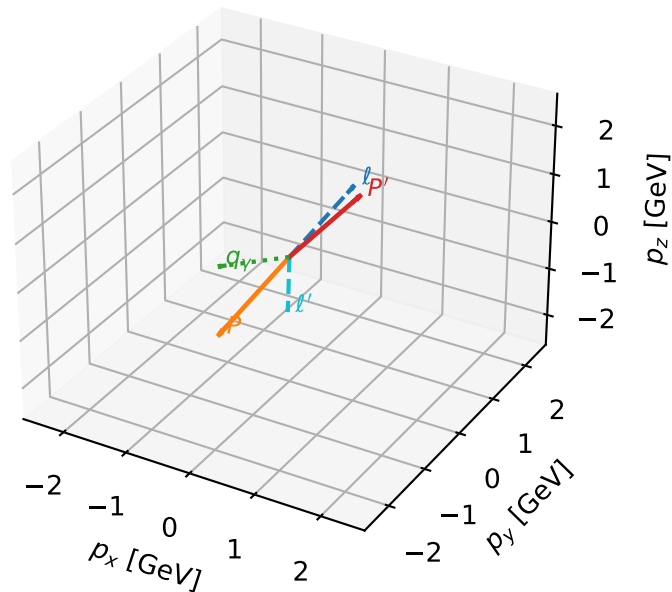
$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton h=+1

Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

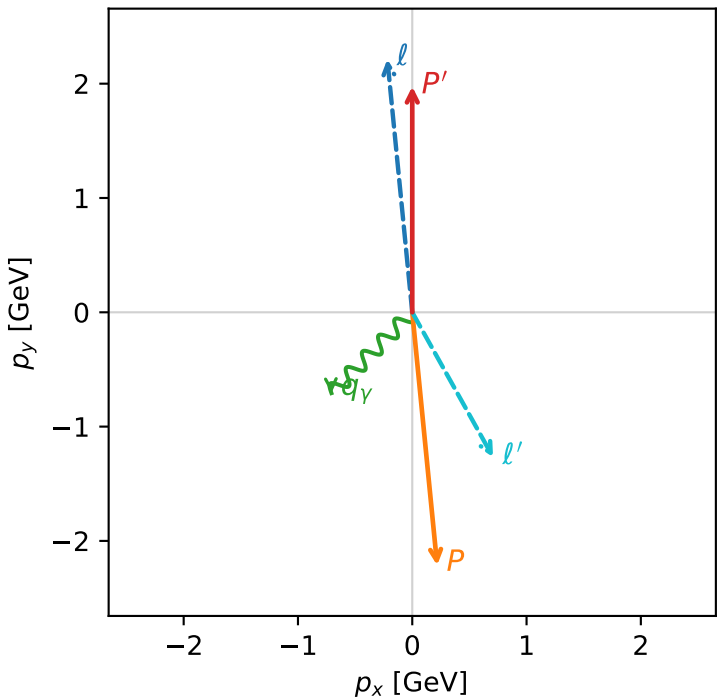


ghz_purity_mid_egamma_l_lepton_region_3 $P_{\text{GHZ+}}=0.383831$
region: mid_Egamma_L_lepton_region_3
outgoing-state purity: 0.998789
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.9752$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.66897$, $\phi_P=4.81056$
 $E_\gamma=1$, $\phi_\gamma=3.92699$
 $Q^2=12.382$, $x_B=0.633397$, $t=-17.5427$, $W^2=8.0464$, $y=0.947304$
 $\theta(\ell', \gamma)=74.1447$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 ℓ' $E= 1.4519$ $p_x= 0.7071$ $p_y= -1.2681$ $p_z= 0.0000$
 P' $E= 2.1866$ $p_x= 0.0000$ $p_y= 1.9752$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7071$ $p_y= -0.7071$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton $h=+1$

$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton $h=+1$

Leading initial-ensemble/final-state components (N=2, retained=99.9%)

